ISM Companion Reader Session 11

Analysis of Variance (ANOVA)

One-way ANOVA, two-way ANOVA without interaction, Sum of Squares, and F-ratio tests.

1 Introduction to Analysis of Variance (ANOVA)

Analysis of Variance (ANOVA) compares means across three or more populations simultaneously. Running multiple two-sample t-tests increases the overall Type I error rate. ANOVA avoids this by testing all means in a single F-ratio test.

Between-Group vs Within-Group Variance

Imagine testing three teaching methods across classrooms. Total variation in test scores comes from two sources:

1. **Between-Group Variation (Treatment):** Differences caused by using different teaching methods.
2. **Within-Group Variation (Error):** Random individual variation among students inside the same classroom.

If between-group variance is significantly larger than within-group noise, the teaching methods produce real differences.

2 One-Way ANOVA Framework

For k treatment groups and total sample size $n = \sum n_i$, total variance splits into:

$$\text{SST} = \text{SSTR} + \text{SSE}$$

where SST is Total Sum of Squares, SSTR is Treatment Sum of Squares, and SSE is Error Sum of Squares.

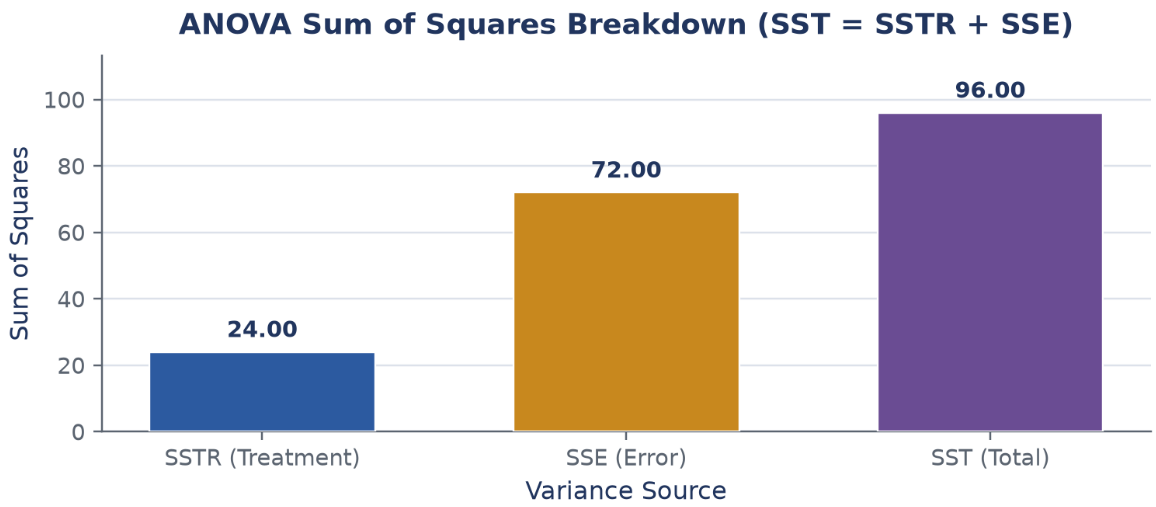

Figure: Breakdown of Total Sum of Squares (SST) into Treatment (SSTR) and Error (SSE).

One-Way ANOVA Table:

Source of Variation	Sum of Squares (SS)	Degrees of Freedom (df)	Mean Square (MS)	F-Ratio
Treatments (Between)	SSTR	$k-1$	$\text{MSTR} = \frac{\text{SSTR}}{k-1}$	$F = \frac{\text{MSTR}}{\text{MSE}}$
Error (Within)	SSE	$n-k$	$\text{MSE} = \frac{\text{SSE}}{n-k}$	–
Total	SST	$n-1$	–	–

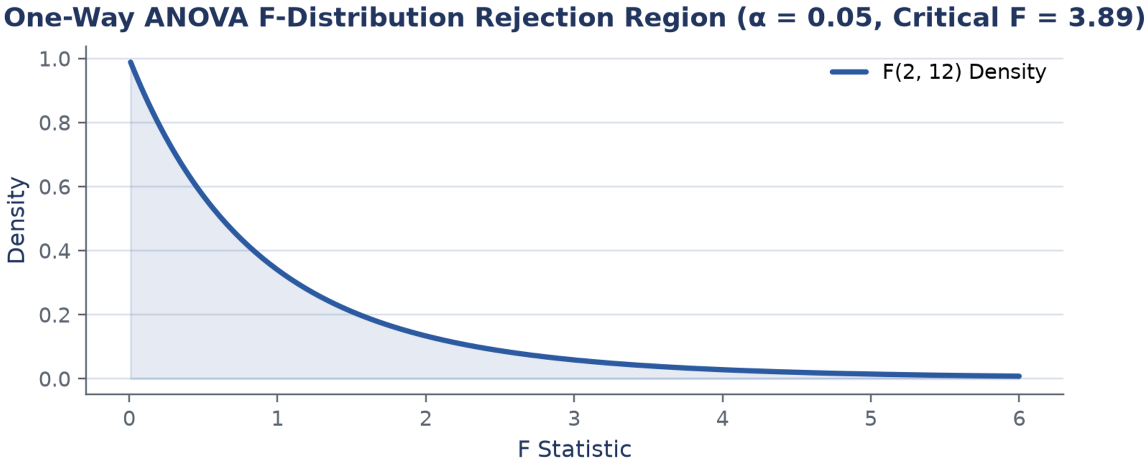

Figure: Fisher F-distribution curve $F(2,12)$ with critical cutoff $F_{0.05} = 3.89$.

Slide Example: One-Way ANOVA Table

Problem Statement: A study compares 3 normal populations with 5 samples each ($k = 3, n = 15$). Given Grand Total $G = 179$, SSTR $= 24.0$, and SSE $= 72.0$, construct the ANOVA table and test for mean equality at $\alpha = 0.05$.

Step 1: Hypotheses: $H_0 : \mu_1 = \mu_2 = \mu_3$ vs H_1 : at least one mean differs.

Step 2: Calculate Degrees of Freedom:

$$df_{\text{treatment}} = k - 1 = 2, \quad df_{\text{error}} = n - k = 12, \quad df_{\text{total}} = 14$$

Step 3: Compute Mean Squares:

$$\text{MSTR} = \frac{24.0}{2} = 12.0, \quad \text{MSE} = \frac{72.0}{12} = 6.0$$

Step 4: Compute F Test Statistic:

$$F_{\text{cal}} = \frac{\text{MSTR}}{\text{MSE}} = \frac{12.0}{6.0} = 2.00$$

Step 5: Decision: Critical value $F_{0.05}(2, 12) = 3.89$. Since $F_{\text{cal}} = 2.00 < 3.89$, we fail to reject H_0. There is no significant difference among the population means.

3 Two-Way ANOVA (Without Interaction)

In Two-Way ANOVA, data is classified by two factors: Treatments (k groups) and Blocks (m blocks).

$$\text{SST} = \text{SSTR} + \text{SSBL} + \text{SSE}$$

Summary Checklist

- **ANOVA Purpose:** Tests equality of $k \geq 3$ population means simultaneously.
- **F-Statistic:** $F = \text{MSTR}/\text{MSE}$. Large F indicates significant treatment differences.

- **Two-Way ANOVA:** Controls for blocking factor, reducing error variance (SSE).